

IGCSE Chemistry CIE

YOUR NOTES



11. Organic Chemistry

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11.1 Formulae, Functional Groups & Terminology

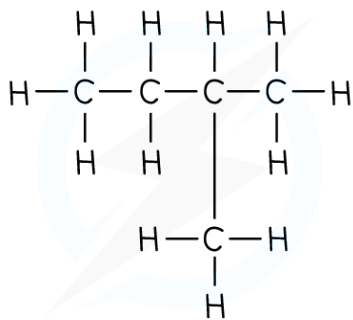
11.1.1 Organic Formulae

Displayed Formulae

- Organic Chemistry is the scientific study of the structure, properties, and reactions of organic compounds.
- Organic compounds are those which contain carbon
- For conventional reasons metal carbonates, carbon dioxide and carbon monoxide are **not** included in organic compounds
- Many of the structures you will be drawing are hydrocarbons
- A **hydrocarbon** is a compound that contains **only** hydrogen and carbon atoms
- Organic compounds can be represented in a number of ways:
 - Displayed Formulae
 - General Formulae
 - Structural Formulae

The **displayed formula** shows the spatial arrangement of all the atoms and bonds in a molecule

- For example:



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- This displayed formula tells us several things about the compound
 - It has 5 carbon atoms
 - It has 12 hydrogen atoms
 - It has only single bonds

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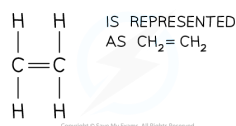
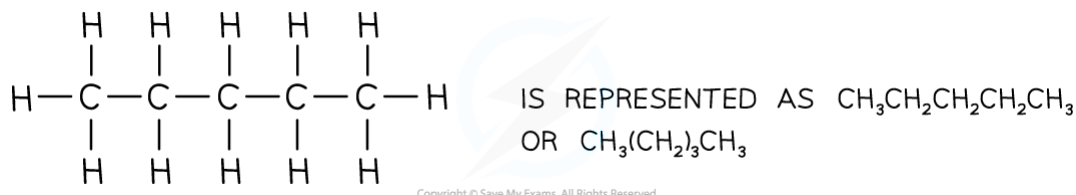
Structural Formulae

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- In structural formulae, enough information is shown to make the structure clear, but most of the actual covalent bonds are omitted
- Only important bonds are always shown, such as double and triple bonds
- Identical groups can be bracketed together
- Side groups are also shown using brackets
- Straight chain alkanes are shown as follows:



Structural Isomers

- Structural isomers are compounds that have the **same molecular** formula but **different structural** formulae
- The **molecular formula** is the **actual number** of atoms of each element in a compound
- Compounds with the same molecular formula can have different structural formulae due to the different arrangement of their atoms in space
- Two examples of structural isomers are shown below

Table showing Structural Isomerism in C_4H_{10}

Molecular formula	C_4H_{10}	C_4H_{10}
Structural formula	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$	$\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_3$
Displayed formula	$ \begin{array}{cccc} \text{H} & \text{H} & \text{H} & \text{H} \\ & & & \\ \text{H}-\text{C} & -\text{C} & -\text{C} & -\text{C}-\text{H} \\ & & & \\ \text{H} & \text{H} & \text{H} & \text{H} \end{array} $	$ \begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\text{H} \\ \\ \begin{array}{ccc} \text{H} & & \text{H} \\ & & \\ \text{H}-\text{C} & -\text{C} & -\text{C}-\text{H} \\ & & \\ \text{H} & \text{H} & \text{H} \end{array} \end{array} $

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Table showing Structural Isomerism in C_4H_8

Molecular formula	C_4H_8	C_4H_8
Structural formula	$CH_3CH_2CH=CH_2$	$CH_3CH=CHCH_3$
Displayed formula	$ \begin{array}{ccccccc} & H & & H & & H & \\ & & & & & & \\ H & -C & - & C & - & C & =C & -H \\ & & & & & & & \\ & H & & H & & & & H \end{array} $	$ \begin{array}{ccccccc} & H & & & & & H \\ & & & & & & \\ H & -C & - & C & = & C & -C & -H \\ & & & & & & & \\ & H & & H & & H & & H \end{array} $

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Exam Tip

Remember: Only double and triple bonds are shown in structural formulae.

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11.1.2 Homologous Series

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Homologous Series

- This is a series or family of organic compounds that have similar features and chemical properties due to them having the same **functional group**
- The functional group is a group of atoms which are bonded in a specific arrangement that is responsible for the characteristic reactions of each member of a homologous series

Table of Compounds & their Functional Groups

Family	Functional group	Name
Alkane	$\begin{array}{c} \quad \\ -C-C- \\ \quad \end{array}$	-ane
Alkene	$\begin{array}{c} \quad \\ -C=C- \\ \quad \end{array}$	-ene
Alcohol	-OH	-anol
Carboxylic acid	$\begin{array}{c} O \\ \\ -C-OH \end{array}$	-anoic acid
Amine	-NH ₂	-amine
Ester	$\begin{array}{c} O \\ \\ -C-O- \end{array}$	-yl-anoate

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Exam Tip

Make sure you can identify the functional group for each homologous series.



General Formulae

General Formulae

- This type of formula tells you the composition of any member of a whole homologous series of organic compound
- For example, all of the alkanes have the general formula C_nH_{2n+2} , where n represents the number of carbon atoms
- This tells you that however many carbon atoms there are in the alkane, doubling this number and adding two will give you the number of hydrogen atoms present in the alkane
- General formulae can be used to work out the formula of a compound from different homologous series if the number of carbon atoms present is known

General Formula of Common Homologous Series

Homologous Series	General Formula
Alkanes	C_nH_{2n+2}
Alkenes	C_nH_{2n}
Alcohols	$C_nH_{2n+1}OH$
Carboxylic Acids	$C_nH_{2n+1}COOH$



Worked Example

What is the formula of an alcohol that contains 5 carbon atoms?

Answer

Number of carbons = 5

Number of hydrogens (excluding in the functional group) = $2 \times 5 + 1 = 11$

Formula = $C_5H_{11}OH$



Worked Example

A compound has the formula $C_{12}H_{24}$. To which homologous series does this compound belong to?

Answer

There are 12 carbon atoms, so $n = 12$

There are twice the number of hydrogen atoms than carbon atoms = $2n$

Therefore the general formula of the compound is C_nH_{2n} which means this compound is an alkene

General Characteristic of Homologous Series

EXTENDED

Characteristics of a Homologous Series

- All members of a homologous series have:
 - The same **general formula**
 - Same **functional group**
 - Similar **chemical properties**
 - Gradation in their physical properties, such as melting and boiling point
 - The difference in the molecular formula between one member and the next is CH_2
 - These characteristics are shown below for ethanol and propanol, which belong to homologous series, alcohols

Table of Characteristics of Ethanol and Propanol

	Ethanol	Propanol
Molecular formula	$\text{CH}_3\text{CH}_2\text{OH}$	$\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
General formula	$\text{C}_n\text{H}_{2n+1}\text{OH}$	$\text{C}_n\text{H}_{2n+1}\text{OH}$
Functional group	$-\text{OH}$	$-\text{OH}$
Boiling point ($^{\circ}\text{C}$)	78	98

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Exam Tip

Make sure you learn the general formula for each homologous series.

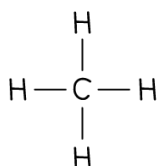
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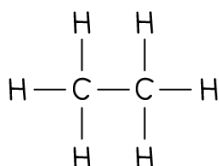
11.1.3 Saturated & Unsaturated Compounds

Saturated & Unsaturated Compounds

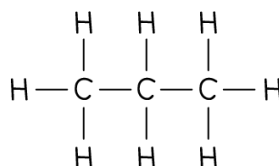
- **Saturated compounds** have molecules in which all carbon-carbon bonds are **single bonds**
- Examples of compounds that are saturated are **alkanes**
- Alkanes are saturated hydrocarbons with the general formula C_nH_{2n+2}



METHANE



ETHANE



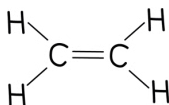
PROPANE

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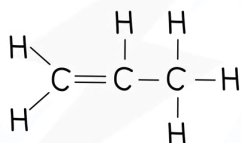
Alkanes contain only carbon-carbon single bonds so are saturated

- **Unsaturated compounds** consist of molecules in which one or more carbon-carbon bonds are **not** single bonds
- They contain carbon-carbon **double bonds** ($C=C$)
- Examples of compounds that are unsaturated are alkenes.
- **Alkenes** are unsaturated hydrocarbons with the general formula is C_nH_{2n}
- The presence of the double bond, $C=C$, means they can **make more bonds** with other atoms by opening up the $C=C$ bond and allowing incoming atoms to form another single bond with each carbon atom of the functional group
- Each of these carbon atoms now forms 4 single bonds instead of 1 double and 2 single bonds

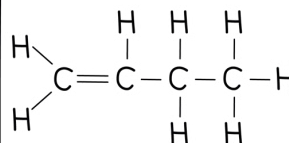
ETHENE



PROPENE



BUT-1-ENE



Alkenes contain one carbon-carbon double bond so are unsaturated



Exam Tip

Remember: Saturated compounds have Single bonds only. Unsaturated compounds have doUble bonds

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11.1.4 Naming Organic Compounds

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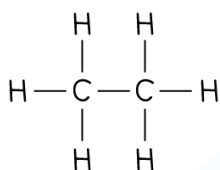


Naming Organic Compounds

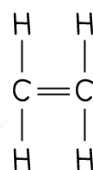
- The names of organic compounds have two parts: the prefix (or stem) and the end part (or suffix)
- The prefix tells you how many carbon atoms are present in the longest continuous chain in the compound
- The suffix tells you what functional group is on the compound

FIRST PART OF NAME		SECOND PART OF NAME		
NAME	NUMBER OF CARBON ATOMS	NAME	FUNCTIONAL GROUP	FAMILY
METH...	1	...ANE	NONE	ALKANE
ETH...	2	...ENE	C=C BOND	ALKENE
PROP...	3	...ANOL	R-OH	ALCOHOL
BUT...	4	...ANOIC ACID	R-C=O-OH	CARBOXYLIC ACID
PENT...	5	...AMINE	R-NH ₂	AMINE
HEX...	6	...YL ...ANOATE	R-C=O-O-R	ESTER

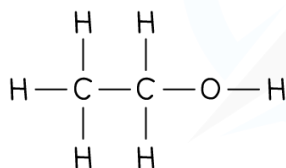
Structures of organic compounds



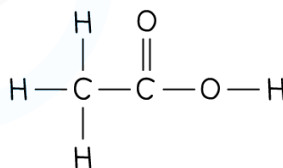
ETHANE



ETHENE



ETHANOL



ETHANOIC ACID

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Exam Tip

Make sure you can draw and name the structures given above.

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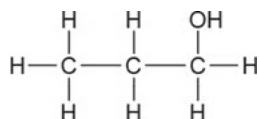


Further Naming of Organic Compounds

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Further Rules for Naming Compounds

- When there is more than one carbon atom where a functional group can be located it is important to distinguish exactly **which carbon** the functional group is on
- Each carbon is numbered and these numbers are used to describe where the functional group is
- For example:
 - Propan-1-ol is alcohol with an -OH functional group
 - The 2 in the name indicates that the -OH group is located on the second carbon atom



In propan-1-ol the -OH group is located on the first carbon atom

Alkanes

Alkane	Structural formula	Displayed formula
Methane	CH ₄	$ \begin{array}{c} \text{H} \\ \\ \text{H} - \text{C} - \text{H} \\ \\ \text{H} \end{array} $
Ethane	CH ₃ CH ₃	$ \begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H} - \text{C} - \text{C} - \text{H} \\ \quad \\ \text{H} \quad \text{H} \end{array} $
Propane	CH ₃ CH ₂ CH ₃	$ \begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \\ \text{H} - \text{C} - \text{C} - \text{C} - \text{H} \\ \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \end{array} $
Butane	CH ₃ CH ₂ CH ₂ CH ₃	$ \begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \quad \\ \text{H} - \text{C} - \text{C} - \text{C} - \text{C} - \text{H} \\ \quad \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \end{array} $

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Alkenes

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Alkene	Structural formula	Displayed formula
Ethene	$\text{CH}_2=\text{CH}_2$	$ \begin{array}{c} \text{H} & & \text{H} \\ & \diagdown & / \\ & \text{C}=\text{C} \\ & / & \diagdown \\ \text{H} & & \text{H} \end{array} $
Propene	$\text{CH}_2=\text{CHCH}_3$	$ \begin{array}{c} \text{H} & & \text{H} & \text{H} \\ & \diagdown & / & \\ & \text{C}=\text{C} & -\text{C}-\text{H} \\ & / & & \\ \text{H} & & & \text{H} \end{array} $
But-1-ene	$\text{CH}_2=\text{CHCH}_2\text{CH}_3$	$ \begin{array}{c} \text{H} & \text{H} & \text{H} & \text{H} \\ & & & \\ \text{C}=\text{C} & -\text{C} & -\text{C}-\text{H} \\ & & & \\ \text{H} & & \text{H} & \text{H} \end{array} $
But-2-ene	$\text{CH}_3\text{CH}=\text{CHCH}_3$	$ \begin{array}{c} \text{H} & \text{H} & \text{H} & \text{H} \\ & & & \\ \text{H}-\text{C} & -\text{C} & =\text{C} & -\text{C}-\text{H} \\ & & & \\ \text{H} & & & \text{H} \end{array} $

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Alcohols

Alcohol	Structural formula	Displayed formula
Methanol	CH_3OH	$ \begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\text{O}-\text{H} \\ \\ \text{H} \end{array} $
Ethanol	$\text{CH}_3\text{CH}_2\text{OH}$	$ \begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}-\text{C}-\text{C}-\text{O}-\text{H} \\ \quad \\ \text{H} \quad \text{H} \end{array} $
Propan-1-ol	$\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$	$ \begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{O}-\text{H} \\ \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \end{array} $
Propan-2-ol	$\text{CH}_3\text{CHOHCH}_3$	$ \begin{array}{c} \text{H} \quad \text{OH} \quad \text{H} \\ \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{H} \\ \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \end{array} $
Butan-1-ol	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$	$ \begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{C}-\text{O}-\text{H} \\ \quad \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \end{array} $
Butan-2-ol	$\text{CH}_3\text{CH}_2\text{CHOHCH}_3$	$ \begin{array}{c} \text{H} \quad \text{H} \quad \quad \text{H} \quad \text{H} \\ \quad \quad \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{OH}-\text{C}-\text{C}-\text{H} \\ \quad \quad \quad \quad \\ \text{H} \quad \text{H} \quad \quad \text{H} \quad \text{H} \end{array} $

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Carboxylic acids

ALKENE	STRUCTURAL FORMULA	DISPLAYED FORMULA
METHANOIC ACID	HCOOH	$\begin{array}{c} \text{O} \\ \\ \text{H}-\text{C}-\text{O}-\text{H} \end{array}$
ETHANOIC ACID	CH_3COOH	$\begin{array}{c} \text{H} \quad \text{O} \\ \quad \\ \text{H}-\text{C}-\text{C}-\text{O}-\text{H} \\ \\ \text{H} \end{array}$
PROPANOIC ACID	$\text{CH}_3\text{CH}_2\text{COOH}$	$\begin{array}{c} \text{H} \quad \text{H} \quad \text{O} \\ \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{O}-\text{H} \\ \quad \\ \text{H} \quad \text{H} \end{array}$
BUTANOIC ACID	$\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$	$\begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \quad \text{O} \\ \quad \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{C}-\text{O}-\text{H} \\ \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \end{array}$

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NAME OF ALCOHOL	NAME OF CARBOXYLIC ACID	NAME OF ESTER
ETHANOL	PROPANOIC ACID	ETHYL PROPANOATE $ \begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}-\text{C}-\text{C}-\text{C}=\text{O} \\ \quad \quad \diagdown \\ \text{H} \quad \text{H} \quad \text{O}-\text{C}-\text{C}-\text{H} \\ \quad \quad \quad \quad \\ \quad \quad \quad \text{H} \quad \text{H} \end{array} $
BUTANOL	METHANOIC ACID	BUTYL METHANOATE $ \begin{array}{c} \text{O} \\ \\ \text{H}-\text{C} \\ \diagdown \\ \text{O}-\text{C}-\text{C}-\text{C}-\text{C}-\text{H} \\ \quad \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \end{array} $
PENTANOL	ETHANOIC ACID	PENTYL ETHANOATE $ \begin{array}{c} \text{H} \quad \text{O} \\ \quad \\ \text{H}-\text{C}-\text{C} \\ \quad \diagdown \\ \text{H} \quad \text{O}-\text{C}-\text{C}-\text{C}-\text{C}-\text{C}-\text{H} \\ \quad \quad \quad \quad \quad \quad \\ \quad \quad \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \end{array} $

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11.2 Organic Families

11.2.1 Fossil Fuels

Common Fossil Fuels

- A fuel is a substance which when burned, releases heat energy
- This heat can be transferred into electricity, which we use in our daily lives
- Most common fossil fuels include coal, natural gas and **hydrocarbons** such as methane and propane which are obtained from crude oil
- Hydrocarbons are made from hydrogen and carbon atoms only
- The main constituent of natural gas is **methane**, CH_4

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Petroleum & Fractional Distillation

Petroleum

- Petroleum is also called **crude oil** and is a complex mixture of hydrocarbons which also contains natural gas
- It is a thick, sticky, black liquid that is found under porous rock (under the ground and under the sea)

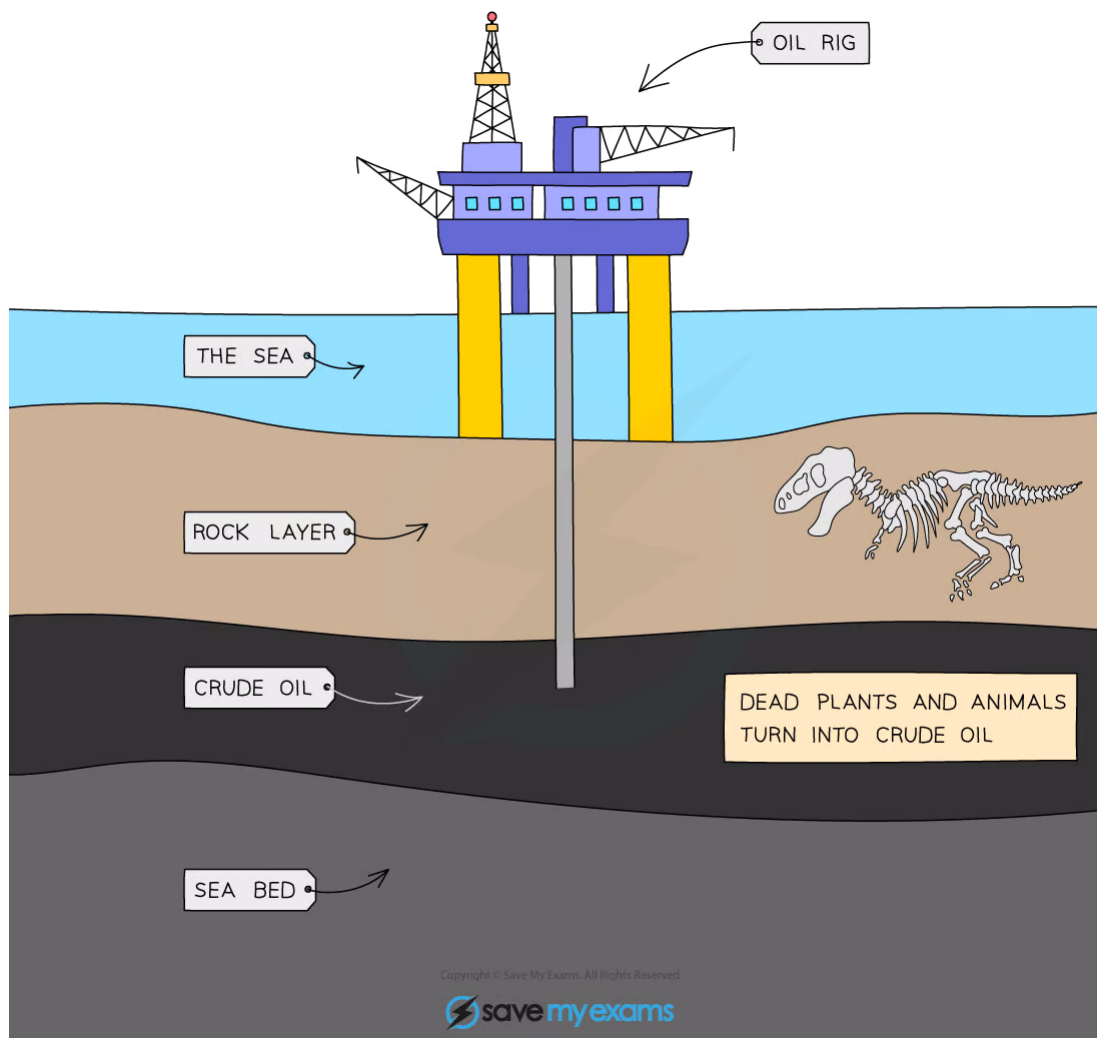


Diagram showing crude oil under the sea

- Petroleum itself as a mixture isn't very useful but each component part of the mixture, called a fraction, is useful and each fraction has different applications
- The fractions in petroleum are separated from each other in a process called **fractional distillation**
- The molecules in each fraction have similar **properties** and **boiling points**, which depend on the number of carbon atoms in the chain

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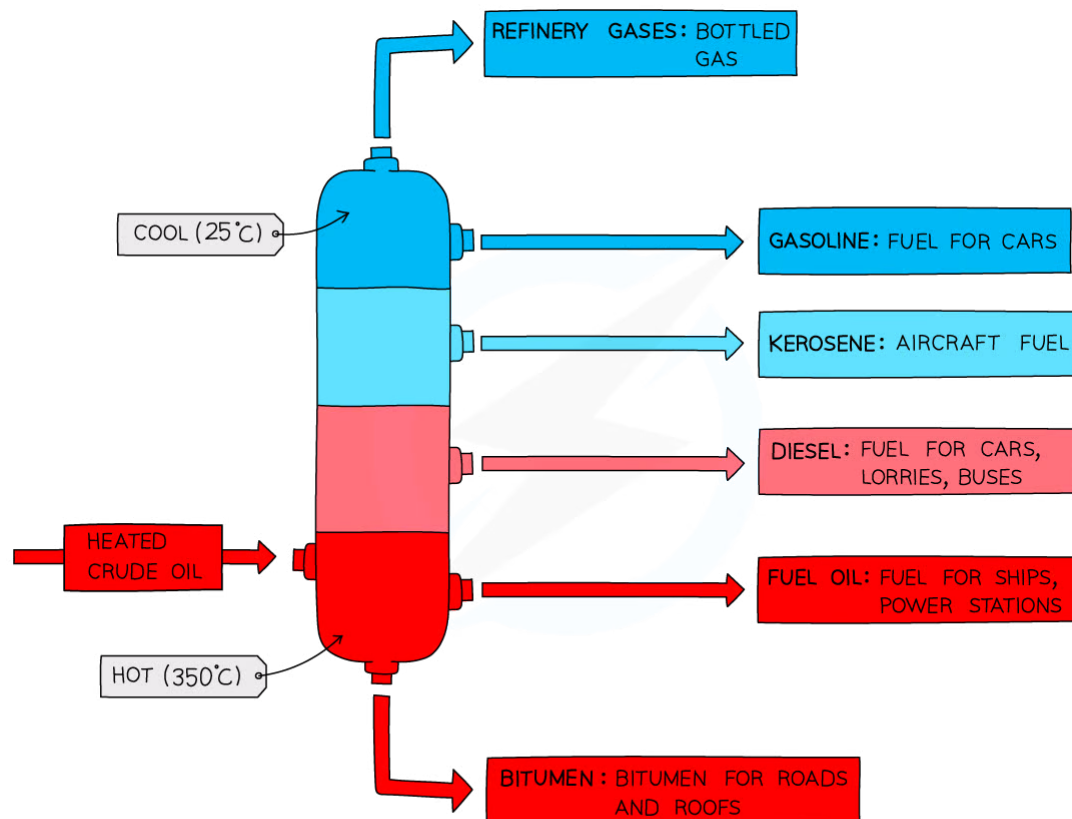


- The **boiling point** and viscosity of each fraction increase as the carbon chain gets longer

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Fractional Distillation



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Diagram showing the process of fractional distillation

- Fractional distillation is carried out in a fractionating column
- The fractionating column is hot at the bottom and cools at the top
- Crude oil enters the fractionating column and is **heated** so **vapours** rise
- Vapours of hydrocarbons with very high boiling points will immediately turn into liquid and are tapped off at the bottom of the column
- Vapours of hydrocarbons with low boiling points will rise up the column and condense at the top to be tapped off
- The different fractions condense at different heights according to their boiling points and are tapped off as liquids.
- The fractions containing smaller hydrocarbons are collected at the top of the fractionating column as gases
- The fractions containing bigger hydrocarbons are collected at the lower sections of the fractionating column

Properties of Fractions

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- **Viscosity**
 - This refers to the ease of flow of a liquid.
 - High viscosity liquids are thick and flow less easily.
 - If the number of carbon atoms increases, the attraction between the hydrocarbon molecules also increases which results in the liquid becoming more viscous with the increasing length of the hydrocarbon chain.
 - The liquid flows less easily with increasing molecular mass
- **Colour**
 - As carbon chain length increases the colour of the liquid gets darker as it gets thicker and more viscous
- **Melting point/boiling point**
 - As the molecules get larger, the intermolecular attraction becomes greater.
 - More heat is needed to separate the molecules.
 - With increasing molecular size there is an increase in boiling point
- **Volatility**
 - Volatility refers to the tendency of a substance to vaporise.
 - With increasing molecular size hydrocarbon liquids become less volatile.
 - This is because the attraction between the molecules increases with increasing molecular size

Uses of Fractions

- Refinery gas: heating and cooking
- Gasoline: fuel for cars (petrol)
- Naphtha: raw product for producing chemicals
- Kerosene: for making jet fuel (paraffin)
- Diesel: fuel for diesel engines (gas oil)
- Fuel oil: fuel for ships and for home heating
- Lubricating oil: for lubricants, polishes, waxes
- Bitumen: for surfacing roads

Trends in Properties

FRACTION	NUMBER OF CARBON ATOMS	BOILING POINT RANGE /°C	BOILING POINT & VISCOSITY INCREASE GOING DOWN
REFINERY GAS	1 – 4	BELOW 25	
GASOLINE / PETROL	4 – 12	40 – 100	
NAPHTHA	7 – 14	90 – 150	
KEROSENE / PARAFFIN	12 – 16	150 – 240	
DIESEL / GAS OIL	14 – 18	220 – 300	
FUEL OIL	19 – 25	250 – 320	
LUBRICATING OIL	20 – 40	300 – 350	
BITUMEN	MORE THAN 70	MORE THAN 350	



Exam Tip

When defining a hydrocarbon, ensure you say that it has hydrogen and carbon atoms **only**.

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11.2.2 Alkanes

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Alkanes: Properties & Bonding

- Alkanes are a group of **saturated** hydrocarbons
- The term saturated means that they only have single carbon-carbon bonds, there are no double bonds
- The general formula of the alkanes is C_nH_{2n+2}
- They are colourless compounds which have a gradual change in their physical properties as the number of carbon atoms in the chain increases
- Alkanes are generally unreactive compounds but they do undergo **combustion** reactions, can be **cracked** into smaller molecules and can react with **halogens** in the presence of light in substitution reactions
- Methane is an alkane and is the major component of **natural gas**
- Methane undergoes complete combustion forming carbon dioxide and water:



This Table shows the Displayed Formula of the First Four Members of the Alkane Homologous Series

Structural Formula	Name	Molecular Formula
<pre> H H - C - H H </pre>	methane	CH_4
<pre> H H H - C - C - H H H </pre>	ethane	C_2H_6
<pre> H H H H - C - C - C - H H H H </pre>	propane	C_3H_8
<pre> H H H H H - C - C - C - C - H H H H H </pre>	butane	C_4H_{10}

Substitution Reaction of Alkanes with Halogens

Extended

- In a **substitution reaction**, one atom (or group of atoms) is swapped with another atom (or group of atoms)
- Alkanes undergo a substitution reaction with halogens in the presence of ultraviolet radiation (sunlight is a source of UV radiation)
- This is called a **photochemical** reaction
- The UV light provides the **activation energy**, E_a , for the reaction
- A hydrogen atom is replaced with the halogen atom
- More than one hydrogen atom can be substituted depending on the amount of ultraviolet radiation there is



In the presence of ultraviolet (UV) radiation, methane reacts with chlorine to form chloromethane and hydrogen chloride



Exam Tip

You need to be able to draw the displayed and structural formulae of the products formed when one halogen atom replaces a hydrogen (also known as monosubstitution)

YOUR NOTES



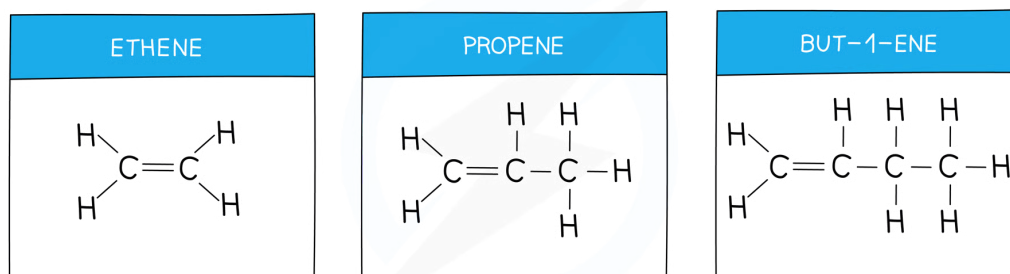
11.2.3 Alkenes

YOUR NOTES



Catalytic Cracking

- Alkenes are unsaturated hydrocarbons with carbon-carbon double bonds (C=C)
- Their general formula is C_nH_{2n}
- The presence of the double bond, C=C, means they can **make more bonds** with other atoms by opening up the C=C bond and allowing incoming atoms to form another single bond with each carbon atom of the functional group
- Each of these carbon atoms now forms 4 single bonds instead of 1 double and 2 single bonds
- This makes them much more reactive than alkanes

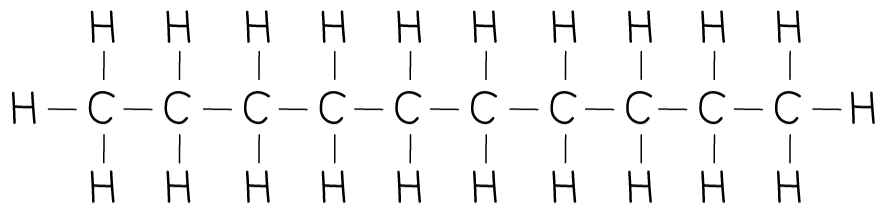


The displayed formula of the first three alkenes

Manufacture of Alkenes

- Although there is use for each fraction obtained from the fractional distillation of crude oil, the amount of longer chain hydrocarbons produced is far greater than needed
- These long chain hydrocarbon molecules are further processed to produce other products
- A process called catalytic cracking is used to convert longer-chain molecules into short-chain and more useful hydrocarbons
- Shorter chain alkanes, alkenes and hydrogen are produced from the cracking of longer chain alkanes
- Alkenes can be used to make polymers and the hydrogen used to make ammonia
- Kerosene and diesel oil are often cracked to produce petrol, other alkenes and hydrogen
- Cracking** involves heating the hydrocarbon molecules to around 600 – 700°C to **vaporise** them
- The vapours then pass over a hot powdered **catalyst** of alumina or silica
- This process breaks covalent bonds in the molecules as they come into contact with the surface of the catalyst, causing **thermal decomposition** reactions
- The molecules are broken up in a random way which produces a mixture of smaller alkanes and alkenes
- Hydrogen and a higher proportion of alkenes are formed at higher temperatures and higher pressure

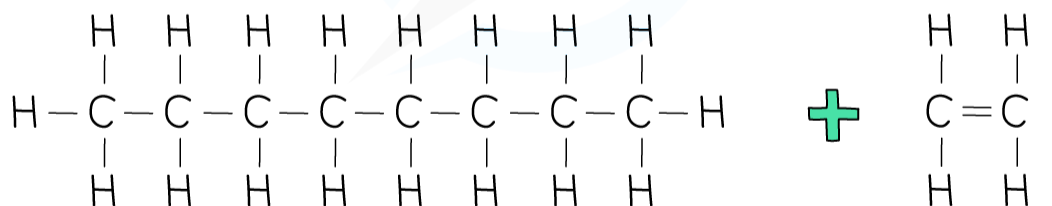
YOUR NOTES



DECANE



HEAT



OCTANE



ETHENE



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The 10 carbon molecule decane is catalytically cracked to produce octane for petrol and ethene for ethanol



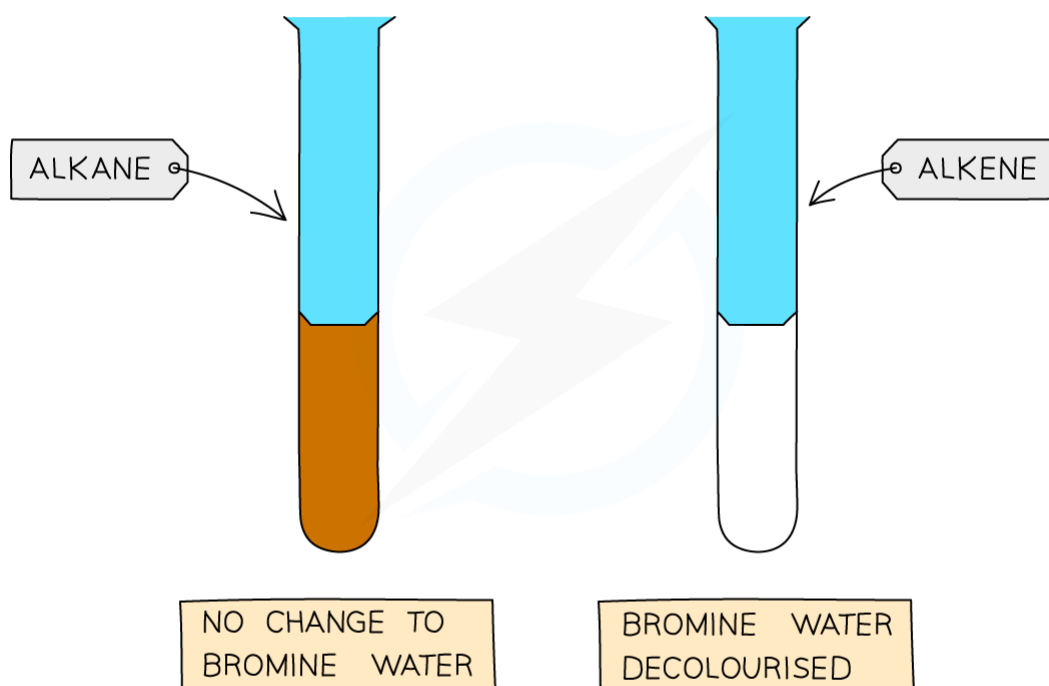
Exam Tip

When describing what happens to bromine water in an alkene ensure you say colourless, and not clear.

Distinguishing from Alkanes

Distinguishing Between Alkanes & Alkenes

- Alkanes and alkenes have different molecular structures
- All alkanes are saturated and alkenes are unsaturated
- The presence of the $C=C$ double bond allows alkenes to react in ways that alkanes cannot
- This allows us to tell alkenes apart from alkanes using a simple chemical test using **bromine water**

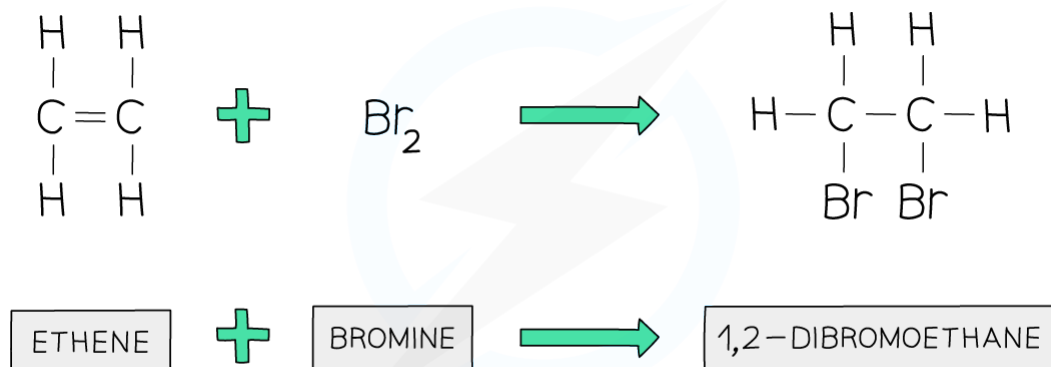


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- Bromine water is an orange coloured solution of bromine
- When bromine water is shaken with an alkane, it will remain as an orange solution as alkanes do not have double carbon bonds ($C=C$) so the bromine remains in solution
- When bromine water is shaken with an alkene, the alkene will decolourise the bromine water and turn colourless as alkenes do have double carbon bonds ($C=C$)
- The bromine atoms add across the $C=C$ double bond hence the solution no longer contains the orange coloured bromine
- This reaction between alkenes and bromine is called an addition reaction

YOUR NOTES





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Each carbon atom of the double bond accepts a bromine atom, causing the bromine solution to lose its colour



Exam Tip

When describing what happens to bromine water in an alkene ensure you say colourless, and not clear.

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11.2.4 Addition Reactions

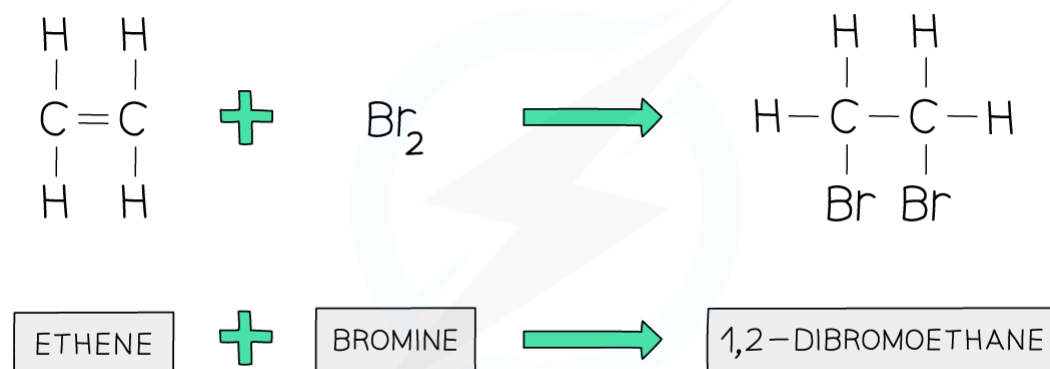
YOUR NOTES



Addition Reactions

EXTENDED

- Alkenes undergo addition reactions in which atoms of a simple molecule add across the C=C double bond
- The reaction between bromine and ethene is an example of an addition reaction

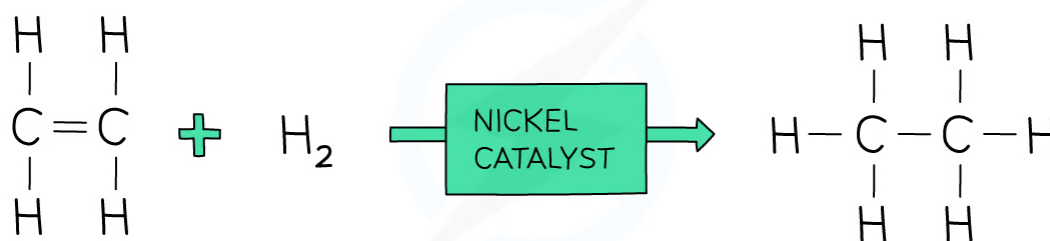


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Bromine atoms add across the C=C in the addition reaction of ethene and bromine

- Alkenes also undergo addition reactions with hydrogen in which an **alkane** is formed
- These are **hydrogenation reactions** and occur at **150°C** using a **nickel catalyst**
- Hydrogenation reactions are used to manufacture margarine from vegetable oils
 - Vegetable oils are polyunsaturated molecules which are partially hydrogenated to increase the M_r and turn the oils into solid fats



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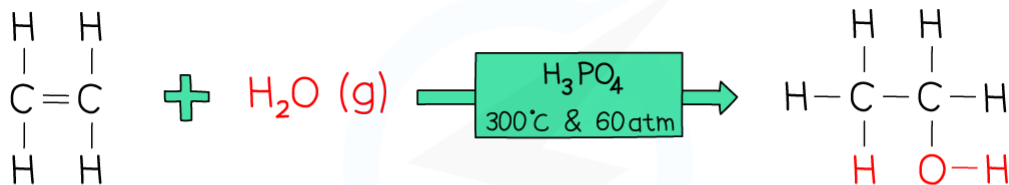


Hydrogen atoms add across the C=C in the hydrogenation of ethene to produce an alkane

- Alkenes also undergo addition reactions with steam in which an **alcohol** is formed.

- Since water is being added to the molecule it is also called a **hydration** reaction
- The reaction is very important industrially for the production of alcohols and it occurs using the following conditions:
 - Temperature of around **300°C**
 - Pressure of **60 – 70 atm**
 - Concentrated phosphoric **acid catalyst**

YOUR NOTES



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A water molecule adds across the C=C in the hydration of ethene to produce ethanol



Exam Tip

You need to be able to draw the displayed formulae of the products of alkenes with water, hydrogen and bromine.

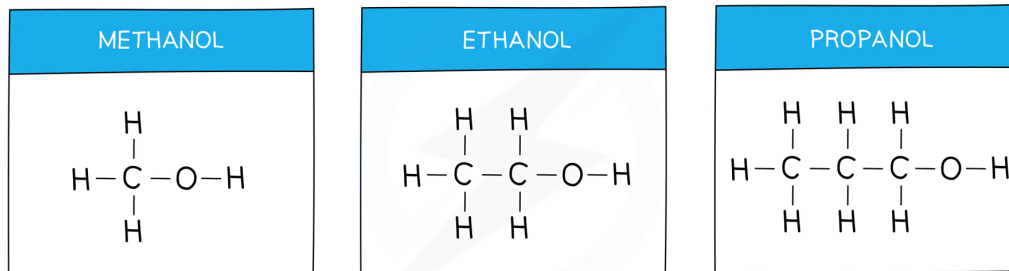
11.2.5 Alcohols

YOUR NOTES



Alcohols

- All alcohols contain the hydroxyl (-OH) functional group which is the part of alcohol molecules that is responsible for their **characteristic reactions**
- Alcohols are a homologous series of compounds that have the general formula $\text{C}_n\text{H}_{2n+1}\text{OH}$
- They differ by one -CH_2 in the molecular formulae from one member to the next

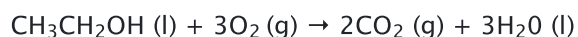


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Diagram showing the first three alcohols

- Ethanol ($\text{C}_2\text{H}_5\text{OH}$) is one of the most important alcohols
- It is the type of alcohol found in **alcoholic drinks** such as wine and beer
- It is also used as fuel for **cars** and as a **solvent**
- Alcohols burn in excess oxygen and produce CO_2 and H_2O
- Ethanol undergoes complete combustion:



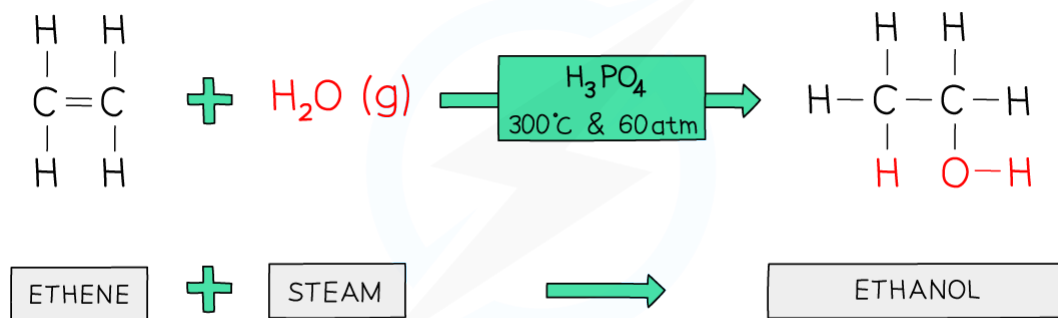
The Manufacture of Ethanol

- There are two methods used to manufacture ethanol:
 - The **hydration** of **ethene** with steam
 - The **fermentation** of glucose
- Both methods have advantages and disadvantages which are considered

Hydration of ethene

- A mixture of ethene and steam is passed over a hot catalyst of **phosphoric acid** at a temperature of approximately **300°C**
- The pressure used is **60 atmospheres (6000kPa)**
- The gaseous ethanol is then condensed into a liquid for use

YOUR NOTES



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A water molecule adds across the C=C in the hydration of ethene to produce ethanol

Fermentation of glucose

- Sugar or starch is dissolved in water and yeast is added
- The mixture is then fermented between **25** and **35 °C** with the **absence** of oxygen for a few days
- Yeast contains enzymes that catalyse the break down of starch or sugar to glucose
- If the temperature is too **low** the reaction rate will be too slow and if it is too **high** the enzymes will become denatured
- The yeast respire anaerobically using the glucose to form ethanol and carbon dioxide:



- The yeast are killed off once the concentration of alcohol reaches around 15%, hence the reaction vessel is emptied and the process is started again
- This is the reason that ethanol production by fermentation is a batch process



Exam Tip

Make sure you learn the conditions for both hydration and fermentation.

Comparing Methods of Ethanol Production

Extended

YOUR NOTES



	HYDRATION OF ETHENE	FERMENTATION
EQUIPMENT	COMPLEX SET-UP REQUIRED	SIMPLE EQUIPMENT NEEDED
RAW MATERIALS	USES NON-RENEWABLE RESOURCES (CRUDE OIL)	USES RENEWABLE RESOURCES (SUGAR CANE)
TYPE OF PROCESS	CONTINUOUS PROCESS – A STREAM OF REACTANT IS CONSTANTLY PASSED OVER THE CATALYST. THIS IS MORE EFFICIENT THAN A BATCH PROCESS	BATCH PROCESS – EVERYTHING IS MIXED TOGETHER IN A REACTION VESSEL AND THEN LEFT FOR SEVERAL DAYS. THAT BATCH IS THEN REMOVED AND A NEW REACTION IS SET UP – THIS IS INEFFICIENT
RATE OF REACTION	FAST	VERY SLOW (SEVERAL DAYS)
QUALITY OF PRODUCT	PRODUCES PURE ETHANOL	PRODUCES A DILUTE SOLUTION REQUIRING FURTHER PROCESSING
ATMOSPHERIC EFFECTS	NO GREENHOUSE GAS IS PRODUCED BUT POLLUTANTS ARE FORMED FROM THE BURNING OF FOSSIL FUELS TO MAINTAIN HIGH TEMPERATURE	CARBON DIOXIDE PRODUCED WHICH IS A GREENHOUSE GAS
REACTION CONDITIONS	HIGH TEMPERATURES AND PRESSURES REQUIRED, INCREASING THE ENERGY INPUT AND COST	LOW TEMPERATURES REQUIRED

11.2.6 Carboxylic Acids

YOUR NOTES

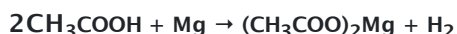


Carboxylic Acids

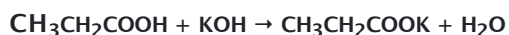
- The carboxylic acids behave like other acids
- They react with:
 - **metals** to form a salt and hydrogen
 - **carbonates** to form a salt, water and carbon dioxide gas
- They also take part in **neutralisation** reactions to produce **salt** and **water**
- **Ethanoic acid** (also called **acetic acid**) is the acid used to make **vinegar**, which contains around 5% by volume ethanoic acid
- The salts formed by the reaction of carboxylic acids all end **-anoate**
- So methanoic acid forms a salt called **methanoate**, **ethanoic** a salt called ethanoate etc.
- In the reaction with metals, a **metal** salt and hydrogen gas are produced

Example reactions of carboxylic acids

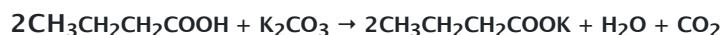
- The reaction of ethanoic acid with **magnesium**, forms the salt magnesium **ethanoate**, and hydrogen gas:



- In the reaction with **hydroxides** a salt and water are formed in a **neutralisation reaction**
- For example the reaction between potassium hydroxide and propanoic acid forms the salt potassium propanoate, and water:



- In the reaction with **carbonates** a metal salt, water and carbon dioxide gas are produced
- For example the reaction between potassium carbonate and butanoic acid, the salt potassium butanoic acid is formed, with water and carbon dioxide



Exam Tip

You need to be able to name and give the formulae of the salts produced in these reactions.

11.2.7 Ethanoic Acid & Esterification Reactions

YOUR NOTES

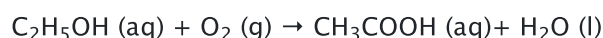


Formation of Ethanoic Acid

EXTENDED

Making Carboxylic Acids

- Two methods used to make carboxylic acids are:
 - Oxidation **by fermentation**
 - Using **oxidising agents**
- The **microbial oxidation** (fermentation) of ethanol will produce a weak solution of vinegar (ethanoic acid)
- This occurs when a bottle of wine is opened as bacteria in the air (acetobacter) will use atmospheric oxygen from air to oxidise the ethanol in the wine



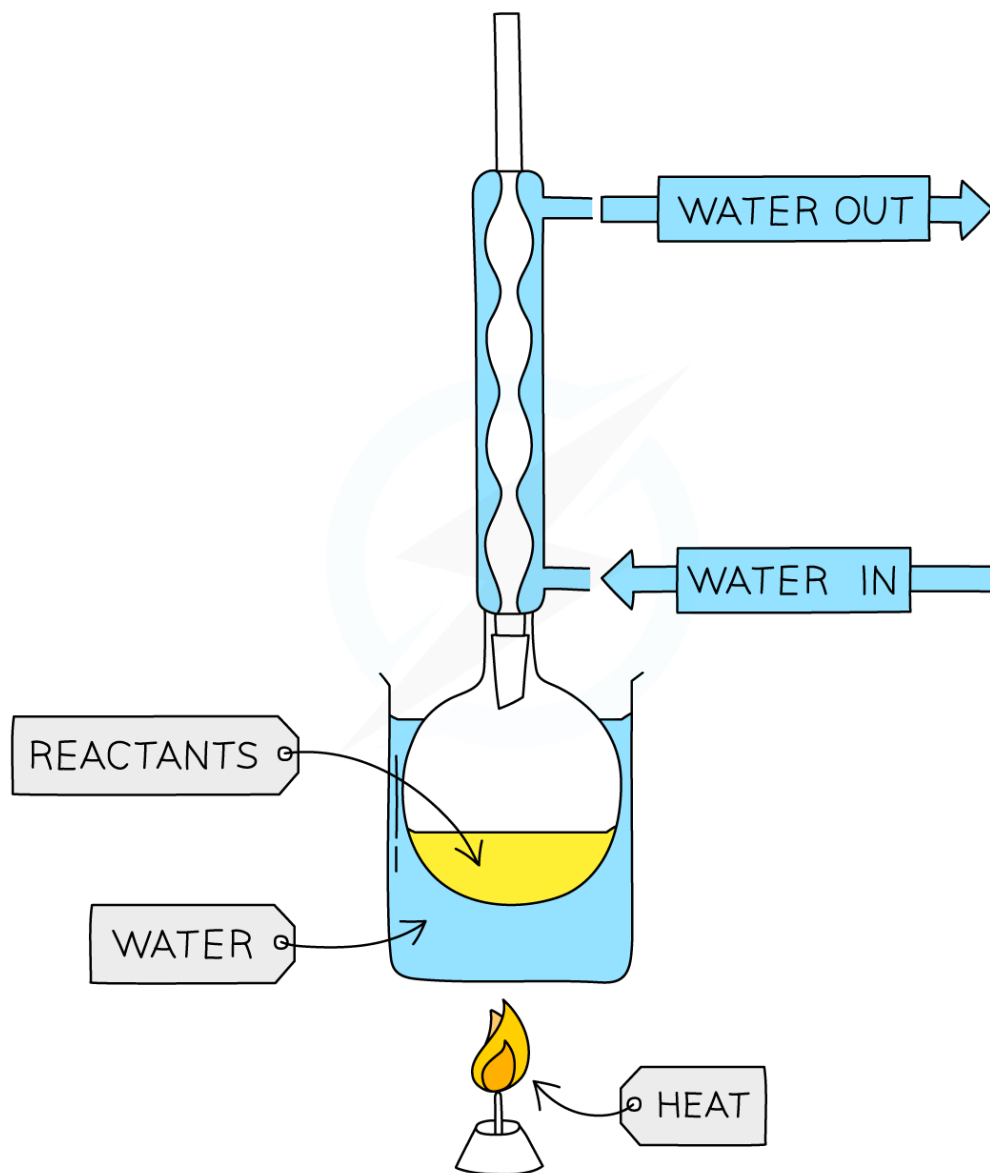
- The acidic, vinegary taste of wine which has been left open for several days is due to the presence of ethanoic acid
- Alternatively, oxidising agent **potassium manganate(VII)** can be used
- This involves heating ethanol with **acidified potassium manganate(VII)** in the presence of an acid
- The heating is performed under reflux which involves heating the reaction mixture in a vessel with a condenser attached to the top
- The condenser prevents the volatile alcohol from escaping the reaction vessel as alcohols have low boiling points
- The equation for the reaction is:



- The solution will change from purple to colourless

- The oxidising agent is represented by the symbol for oxygen in square brackets

YOUR NOTES



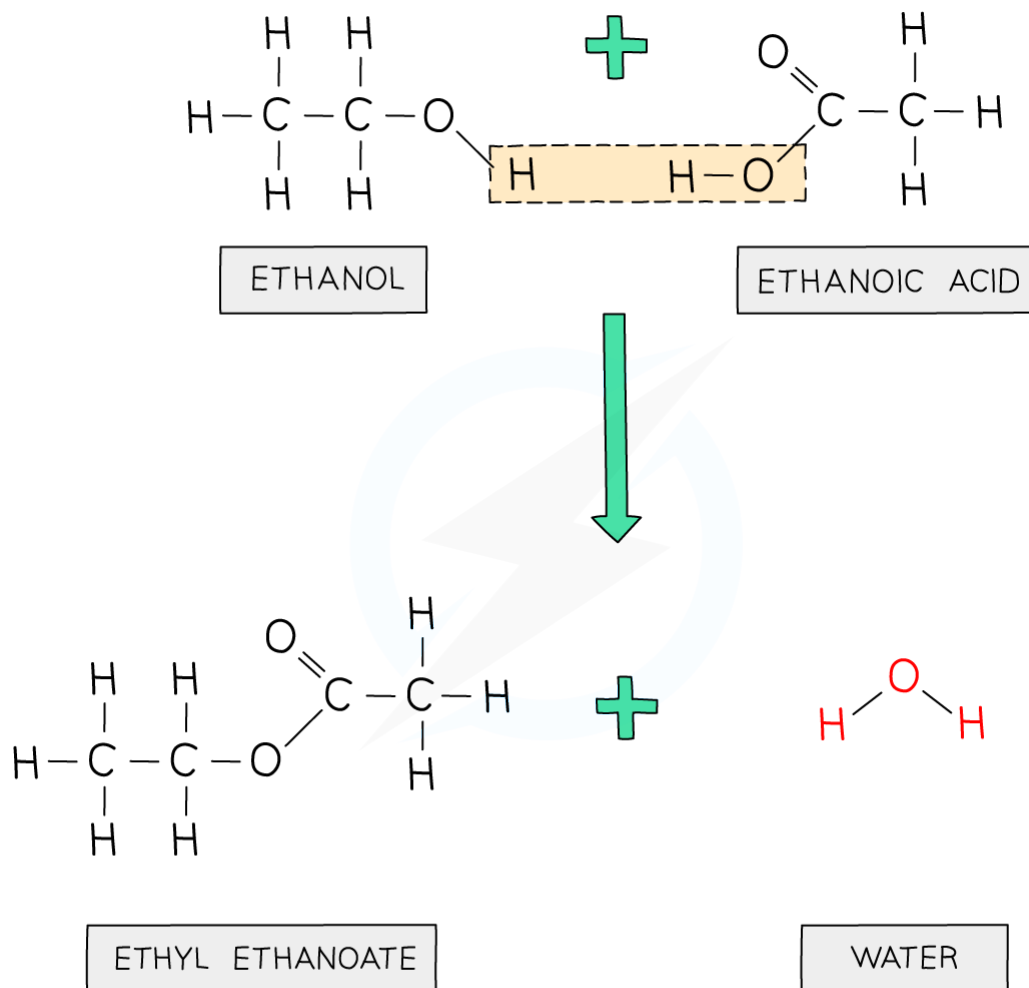
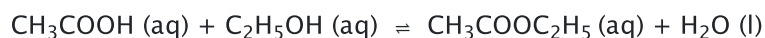
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Diagram showing the experimental setup for the oxidation with KMnO_4 using reflux apparatus

Esterification

EXTENDED

- Alcohols and carboxylic acids react to make esters in **esterification** reactions
- Esters are compounds with the functional group R-COO-R
- Esters are sweet-smelling oily liquids used in food flavourings and perfumes
- Ethanoic acid will react with ethanol in the presence of concentrated sulfuric acid (catalyst) to form ethyl ethanoate:



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Diagram showing the formation of ethyl ethanoate

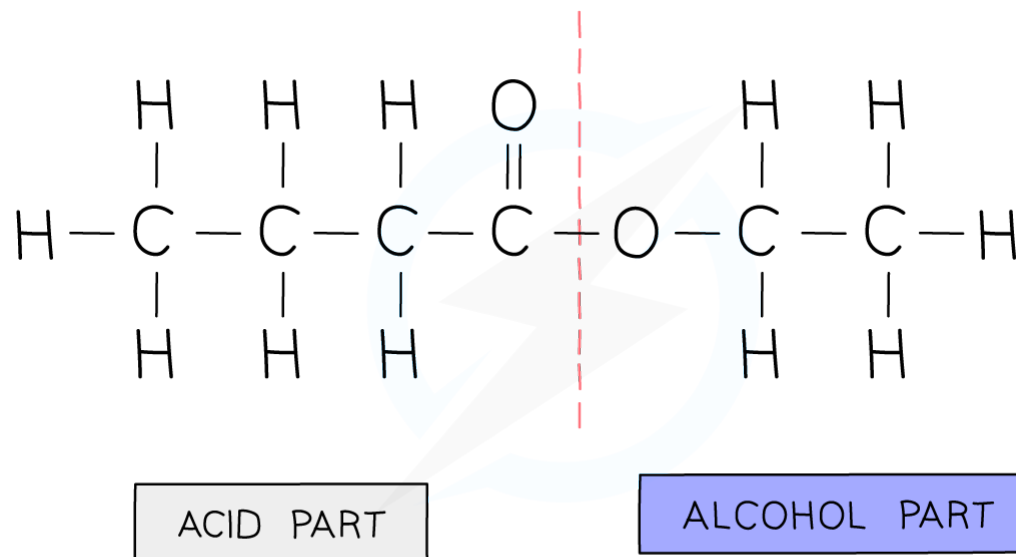
Naming Esters

- An ester is made from an alcohol and carboxylic acid
- The first part of the name indicates the length of the carbon chain in the alcohol, and it ends with the letters '-yl'

YOUR NOTES



- The second part of the name indicates the length of the carbon chain in the carboxylic acid, and it ends with the letters ‘- oate’
 - E.g. the ester formed from **pentanol** and **butanoic acid** is called **pentyl butanoate**



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Diagram showing the origin of each carbon chain in ester; this ester is ethyl butanoate

Table showing the Formation of Esters

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NAME OF ALCOHOL	NAME OF CARBOXYLIC ACID	NAME OF ESTER
ETHANOL	PROPANOIC ACID	ETHYL PROPANOATE $ \begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}-\text{C}-\text{C}-\text{C}=\text{O} \\ \quad \quad \diagdown \\ \text{H} \quad \text{H} \quad \text{O}-\text{C}-\text{C}-\text{H} \\ \quad \quad \quad \quad \\ \quad \quad \quad \text{H} \quad \text{H} \end{array} $
BUTANOL	METHANOIC ACID	BUTYL METHANOATE $ \begin{array}{c} \text{O} \\ \\ \text{H}-\text{C} \\ \diagdown \\ \text{O}-\text{C}-\text{C}-\text{C}-\text{C}-\text{H} \\ \quad \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \end{array} $
PENTANOL	ETHANOIC ACID	PENTYLETHANOATE $ \begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\text{C}=\text{O} \\ \quad \diagdown \\ \text{H} \quad \text{O}-\text{C}-\text{C}-\text{C}-\text{C}-\text{C}-\text{H} \\ \quad \quad \quad \quad \quad \quad \\ \quad \quad \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \end{array} $

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YOUR NOTES



11.3 Polymers

11.3.1 Polymers

Polymers: The Basics

- Polymers are large molecules built by linking 50 or more smaller molecules called monomers
- Each repeat unit is connected to the adjacent units via **covalent bonds**
- Some polymers contain just one type of unit
 - Examples include poly(ethene) and poly(chloroethene), commonly known as PVC
- Others contain two or more different types of monomer units and which are called **copolymers**
 - Examples include nylon and biological proteins
- Different **linkages** also exist, depending on the monomers and the type of polymerisation
 - Examples of linkages are covalent bonds, amide links and ester links

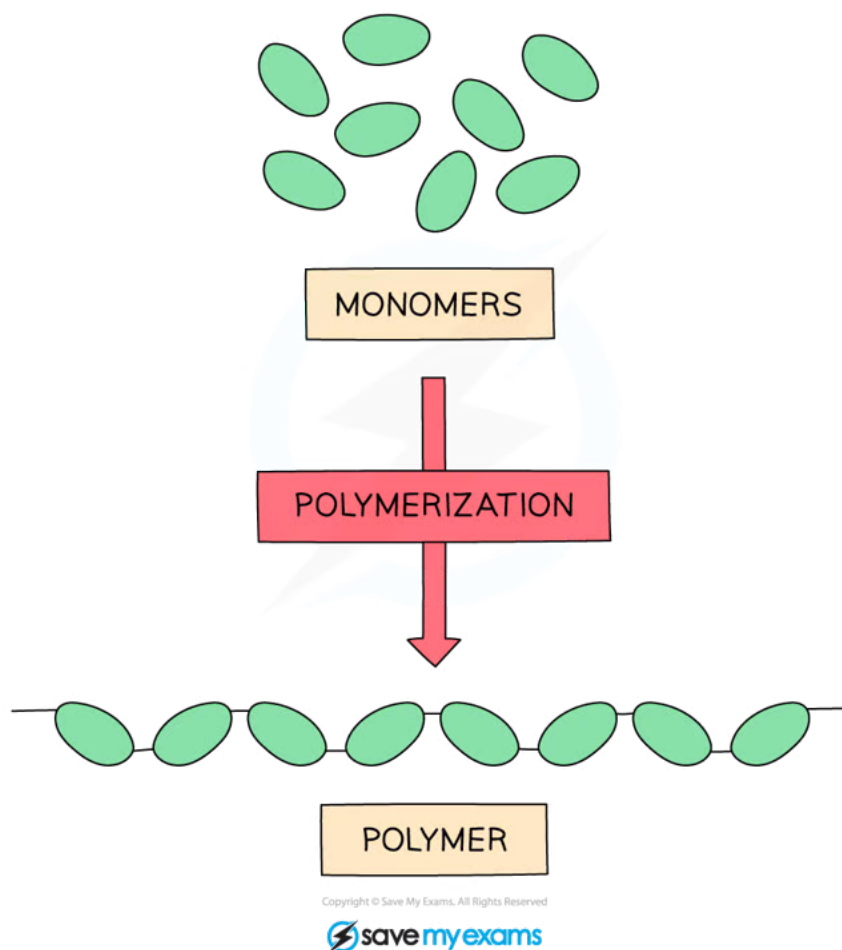
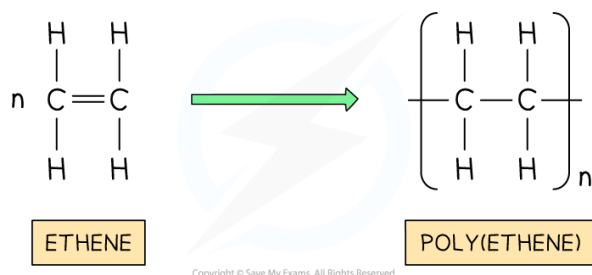


Diagram showing how lots of monomers bond together to form a polymer

YOUR NOTES



- Poly(ethene) is formed by the addition polymerisation of ethene monomers
- Addition polymerisation involves the addition of many monomers to make a long chained polymer
- In this case, many ethene monomers join together due to the carbon carbon double bond breaking



Poly(ethene) is formed by addition polymerisation using ethene monomers

YOUR NOTES



11.3.2 Addition & Condensation Polymers

YOUR NOTES

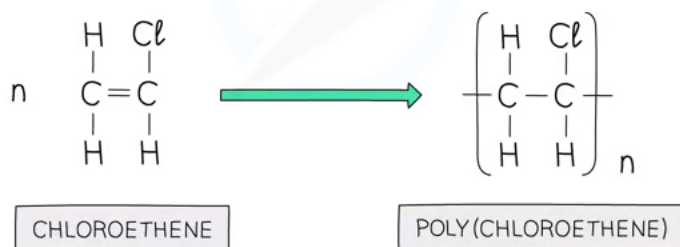
**Addition Polymers****EXTENDED**

- Addition polymers are formed by the joining up of many monomers and only occur in monomers that contain **C=C bonds**
- **One** of the bonds in each C=C bond breaks and forms a bond with the adjacent monomer with the polymer being formed containing **single bonds** only
- Many polymers can be made by the addition of alkene monomers
- Others are made from alkene monomers with different atoms attached to the monomer such as chlorine or a hydroxyl group
- The name of the polymer is deduced by putting the name of the **monomer** in brackets and adding poly- as the **prefix**
- For example if propene is the alkene monomer used, then the name is **poly(propene)**
- Poly(ethene) is formed by the addition polymerisation of **ethene monomers**

Deducing the polymer from the monomer

- Polymer molecules are very large compared with most other molecule
- **Repeat units** are used when displaying the formula
- To draw a repeat unit, change the double bond in the monomer to a **single bond** in the repeat unit
- Add a bond to each end of the repeat unit
- The bonds on either side of the polymer must **extend** outside the brackets (these are called extension or continuation bonds)
- A small **subscript** n is written on the bottom right hand side to indicate a large number of repeat units
- Add on the rest of the groups in the **same order** that they surrounded the double bond in the monomer

YOUR NOTES

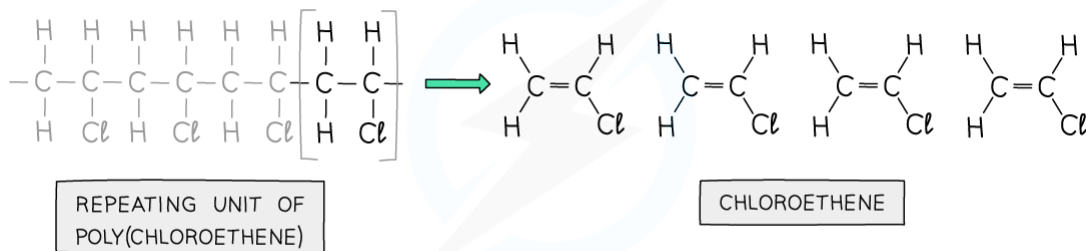


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Examples of addition polymerisation: polythene and PVC

• **Deducing the monomer from the polymer**

- Identify the repeating unit in the polymer
- Change the single bond in the repeat unit to a **double bond** in the monomer
- Remove the bond from each end of the repeat unit



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Diagram showing the monomer from the repeat unit of an addition polymer (polychloroethene)



Exam Tip

You should be able to draw the **box diagrams** representing polymers where each box represents a part of the repeating hydrocarbon chain. The **functional groups** on the monomers and the **link** formed in the polymers are the important parts and must be clearly drawn.

Condensation Polymers

YOUR NOTES

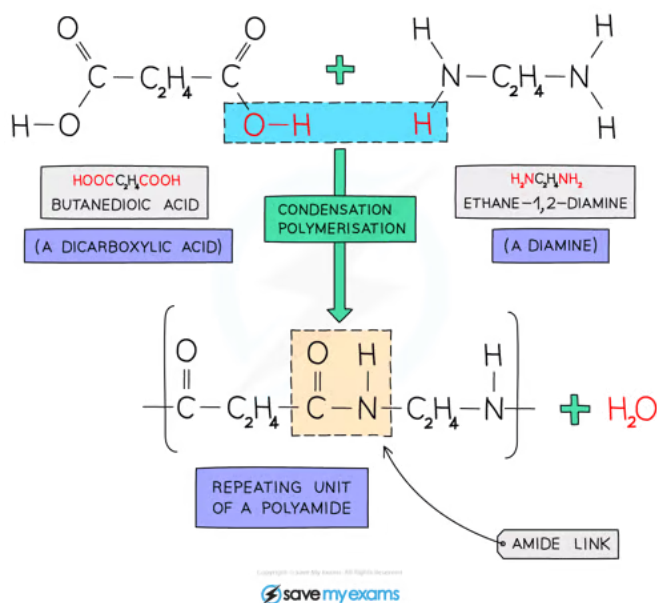


EXTENDED

- Condensation polymers are formed when **two different monomers** are linked together with the **removal** of a small molecule, usually **water**
- This is a key difference between condensation polymers and addition polymers:
 - Addition polymerisation forms the polymer molecule **only**
 - Condensation polymerisation forms the **polymer** molecule and one **water** molecule **per linkage**
- The monomers have two **functional** groups present, one on **each end**
- The functional groups at the ends of one monomer react with the functional group on the end of the other monomer, in so doing creating long chains of **alternating monomers**, forming the polymer
- Hydrolysing (adding water) to the compound in acidic conditions usually reverses the reaction and produces the monomers by rupturing the peptide link

Forming Nylon

- Nylon is a polyamide made from **dicarboxylic acid** monomers (a carboxylic with a -COOH group at **either** end) and **diamines** (an amine with an -NH_2 group at **either** end)
- Each -COOH group reacts with another -NH_2 group on another monomer
- An amide linkage is formed with the subsequent loss of **one** water molecule per link



The condensation reaction in which the polyamide, nylon is produced

- The structure of nylon can be represented by drawing out the polymer using boxes to represent the carbon chains

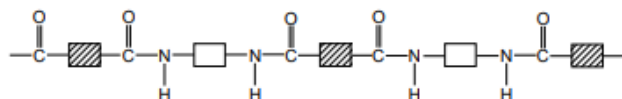
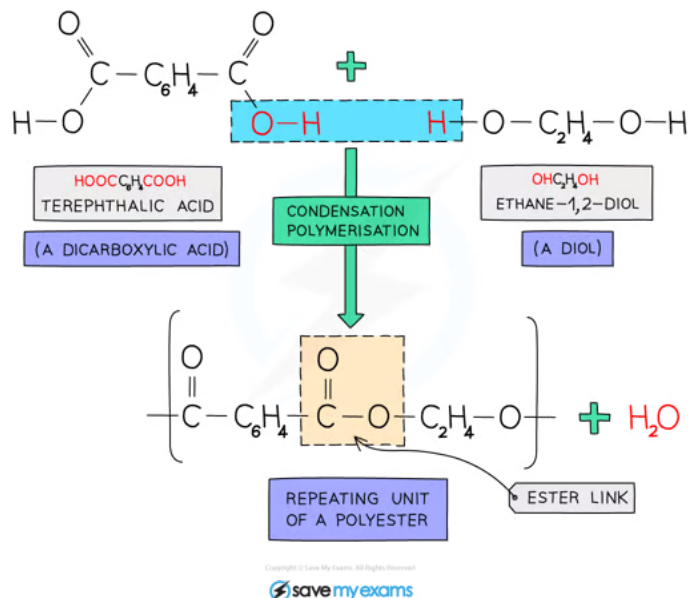


Diagram showing a section of nylon

Forming Polyesters

- PET or polyethylene terephthalate to give its full name, is a **polyester** made from **dicarboxylic acid** monomers (a carboxylic with a -COOH group at **either** end) and **diols** (alcohol with an -OH group at **either** end)
- Each -COOH group reacts with another -OH group on another monomer
- An ester linkage is formed with the subsequent loss of **one** water molecule per link
- For every ester linkage formed in condensation polymerisation, one molecule of water is formed from the combination of a **proton** (H^+) and a **hydroxyl ion** (OH^-)
- PET is also used in synthetic fibres as is sold under the trade name of **terylene**



The condensation reaction in which PET is produced

- The structure of PET can be represented by drawing out the polymer using boxes to represent the carbon chains
- This can be done for all polyesters



Diagram showing a section of PET



Exam Tip

You don't need to know the detailed chemical structure of PET, just the symbolic drawing showing the alternating blocks and the linking ester group. Be careful not to exactly repeat the linking group in nylon or PET; the link alternates by reversing the order of the atoms, rather like a mirror image.

YOUR NOTES



11.3.3 Plastics & their Disposal

YOUR NOTES



Plastics & their Disposal

- **Synthetic** polymers are ones made in a factory, for example nylon, terylene and lycra
- Nylon is a polyamide used to produce **clothing**, **fabrics**, **nets** and **ropes**
- PET, also known as Terylene, is a polyester made from monomers which are joined together by **ester links**
- PET is used extensively in the **textile** industry and is often mixed with cotton to produce **clothing**

Table showing Uses of Plastics

POLYMER	REPEAT UNIT	USES
POLY(ETHENE) i.e. POLYTHENE	$\left[\begin{array}{cc} \text{H} & \text{H} \\ & \\ -\text{C} & -\text{C}- \\ & \\ \text{H} & \text{H} \end{array} \right]_n$	PLASTIC BAGS (LOW DENSITY POLYTHENE) PLASTIC BOTTLES (HIGH DENSITY POLYTHENE)
POLY(PROPENE)	$\left[\begin{array}{cc} \text{CH}_3 & \text{H} \\ & \\ -\text{C} & -\text{C}- \\ & \\ \text{H} & \text{H} \end{array} \right]_n$	FOOD PACKAGING ROPES CARPETS
POLY(CHLOROETHENE) i.e. PVC	$\left[\begin{array}{cc} \text{H} & \text{Cl} \\ & \\ -\text{C} & -\text{C}- \\ & \\ \text{H} & \text{H} \end{array} \right]_n$	PLASTIC SHEETS ARTIFICIAL LEATHER DRAINPIPES AND GUTTERS INSULATION ON WIRES

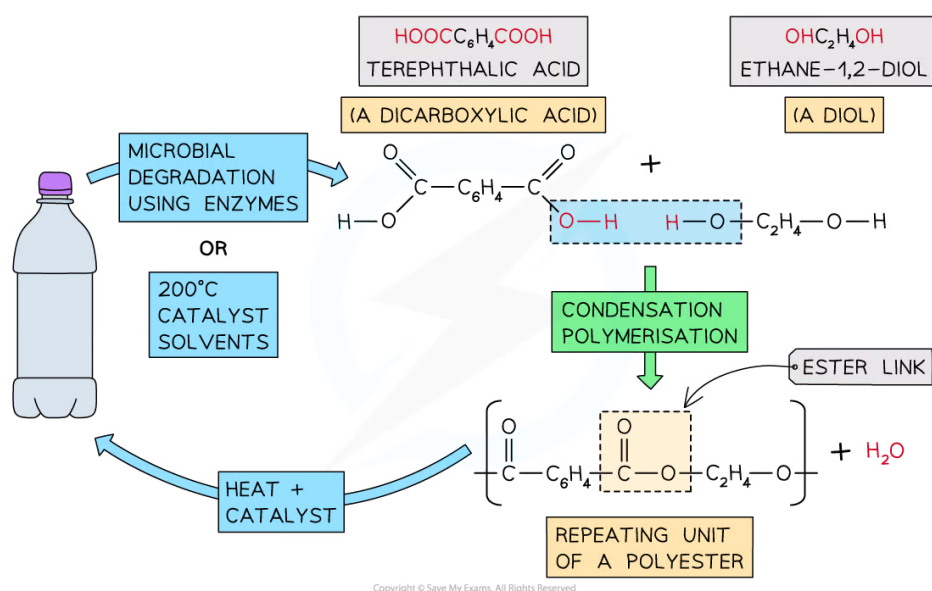
Non-biodegradable plastics

- These are plastics which do not degrade over time or take a very long time to degrade, and cause significant pollution problems
- In particular plastic waste has been spilling over into the **seas** and **oceans** and is causing huge disruptions to marine life
- In landfills waste polymers take up valuable space as they are non-biodegradable so microorganisms cannot break them down. This causes the landfill sites to quickly fill up
- Polymers release a lot of heat energy when **incinerated** and produce **carbon dioxide** which is a greenhouse gas that contributes to climate change
- If incinerated by incomplete combustion, **carbon monoxide** will be produced which is a toxic gas that reduces the capacity of the blood to carry oxygen
- Polymers can be recycled but different polymers must be separated from each other which is a difficult and expensive process

PET Re-polymerisation

- PET stands for polyethylene terephthalate, a common polymer used to make things like plastic bottles
- It is a condensation polymer consisting of repeating ester units, so it is type of polyester, like terylene
- One of the problems with recycling polymers is that the condition needed to break them down, which are usually high temperatures and pressures, can degrade the monomers making them unusable for re-polymerisation
- PET is relatively easy to convert back into the monomers
- It can be depolymerised either using enzymes or by chemical methods
- Enzymes present in microbes breakdown the PET into the original monomers
- The same can be achieved using solvents a catalyst and mild heating

YOUR NOTES



The breakdown of PET into its two monomers takes place using enzymes or chemical catalysts and mild conditions

- The monomers are recovered and be be polymerised into new PET
- This saves on resources and energy, reducing the carbon footprint of the production process

11.3.4 Proteins

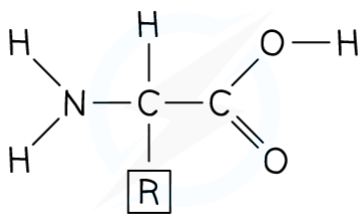
YOUR NOTES



Proteins

Extended

- Proteins are **condensation polymers** which are formed from amino acid monomers joined together by **amide** links (in proteins also known as a peptide link) similar to the structure in nylon
- The units in proteins are different however, consisting of amino acids
- Amino acids are small molecules containing **NH₂** and **COOH** functional groups

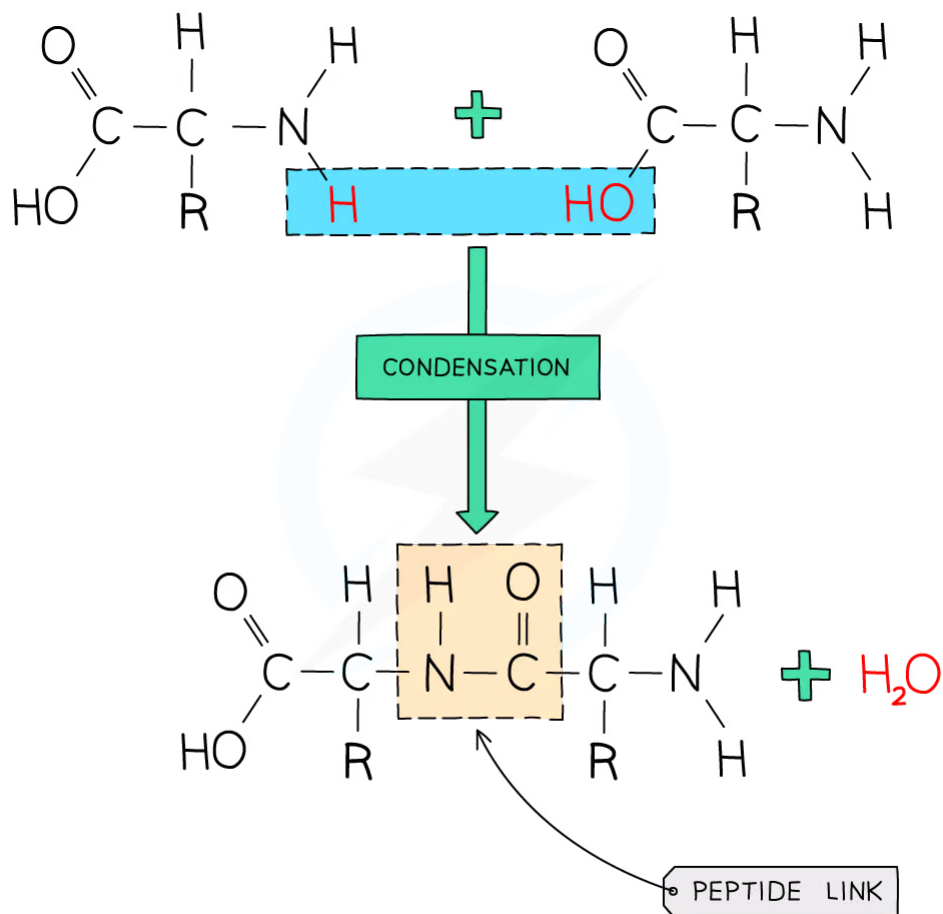


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General structure of an amino acid

- There are **twenty** common amino acids, each differing by their side chain, represented by R
- Proteins can contain between 60 and 600 of these amino acids in different orders
- These are the monomers which polymerise to form the protein

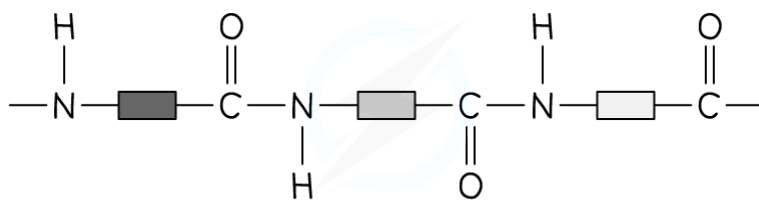
YOUR NOTES



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Diagram showing condensation polymerisation to produce a protein

- The structure of proteins can be represented using the following diagram whereby the boxes represent the carbon chains



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Diagram showing a section of protein